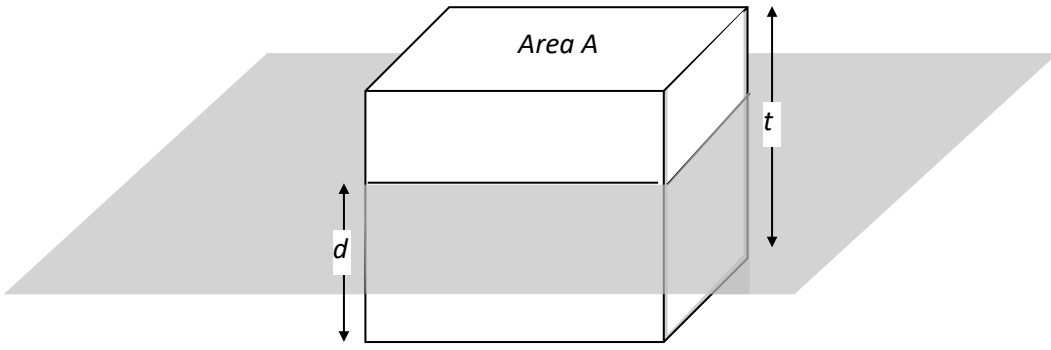
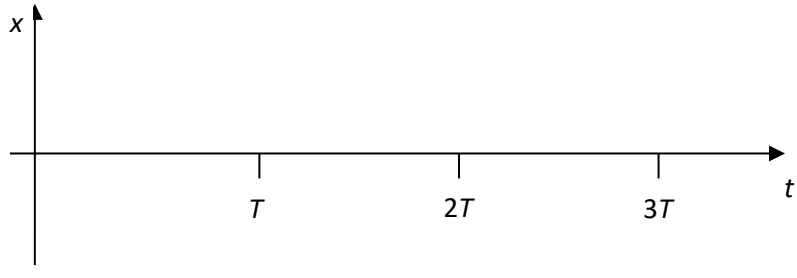
3(a) A rectangular block of wood of cross-section A and thickness t floats horizontally in water as shown in the figure below.



When the block is pushed downwards and then released, it undergoes ***damped*** harmonic motion.

(i) Explain why the motion of the block undergoes damping. [1]

(ii) Sketch a labelled graph showing the variation of displacement x with time t for a time interval over three periods. [2]



(b) Surface water waves from a constant amplitude wave generator are incident on the block. These causes ***forced oscillations*** in the motion of the block. The frequency of the wave generator is varied and ***resonance*** was observed at a particular frequency.

(i) Explain the terms ***'forced oscillations'*** and ***'resonance'*** with reference to the motion of the block. [2]

(ii) Resonance occurs when the water waves incident on the block has a speed of 0.90 m s^{-1} and wavelength 0.30 m.

1. Calculate the frequency of the water waves. [1]

2. Given that the expression for the natural frequency of the oscillating block is given by $f = \frac{1}{2\pi}\sqrt{\frac{28}{m}}$ where f is in Hz and m in kg, calculate the mass of the block. [1]

3. Describe and explain what happens to the amplitude of vertical oscillations of the block after the block has absorbed some water. [2]